Encephalization in the Abyss: Environmental Constraints on Teleost Brain Allometry

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Abstract

Evolutionary marine biologists are keenly interested in what environmental constraints determine the evolutionary pathway of aquatic species. A variable of particular interest to these biologists is brain size, as neural tissue is energetically expensive to maintain but plays a large role in species survival. One might expect that bigger brains are always better, but in the ocean, scientists investigate whether extreme depths, temperature gradients, and predatory food webs actually restrict or alter brain growth.

This project investigates the relationships between brain size and various environmental and ecological characteristics among species of teleost fishes. Using data on average brain mass, body mass, maximum habitat depth, maximum absolute latitude, and trophic level, the study explores how these variables are interrelated and what they reveal about marine evolution. Through exploratory data analysis and statistical modelling, the strength of association between brain mass and the other variables is assessed.

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1 Introduction

Understanding the factors that influence the evolution of brain size in marine teleosts is a central question in evolutionary biology. While brain size is widely regarded as a key indicator of cognitive ability, behavioural complexity, and adaptive success, neural tissue is notoriously expensive to maintain metabolically. In the ocean, particularly at extreme depths, species face severe energetic constraints due to low temperatures, high pressure, and food scarcity. Consequently, biologists have sought to determine how harsh environmental and ecological characteristics dictate or restrict brain development.

In this study, data from a diverse array of teleost species are analyzed to explore possible associations between brain mass and four other biological and environmental variables: body mass, maximum habitat depth, maximum absolute latitude, and trophic level. These variables represent essential aspects of a marine species' ecological and life-history strategy—body mass reflects fundamental allometric scaling and baseline metabolic constraints, maximum depth indicates the physical and energetic pressures of the pelagic and abyssal zones, maximum absolute latitude serves as a proxy for oceanic temperature gradients that dictate metabolic rates, while trophic level reflects the cognitive demands of predatory behaviour versus basic foraging. Examining how these traits vary with brain size can provide valuable insight into evolutionary trade-offs and adaptive strategies across different oceanic zones.

The dataset, stemming from research on encephalization in marine fishes, includes average values for each variable across multiple species. The analysis focuses on exploring relationships between these environmental variables and brain mass, controlling for overall body size, and interpreting the biological implications of observed patterns. By assessing how relative brain size relates to depth, temperature gradients, and predatory status, this study aims to contribute to a deeper understanding of the evolutionary pathways that have shaped teleost diversity in the world's oceans.

2 Data Description

The dataset used in this study is derived from research on the evolutionary ecology of marine fishes, specifically sourced from the compilation by Iglesias et al. ("Life in the unthinking depths") in 2015. It contains information on a diverse array of teleost species. For each species, five key characteristics were recorded and utilized in this analysis:

- **Brain Mass (g):** Average brain mass of the species in grams.

- **Body Mass (g):** Average body mass of the species in grams.
- **Maximum Depth (m):** The maximum recorded habitat depth for the species in meters, representing the physical and hydrostatic pressures of its environment.
- **Maximum Absolute Latitude:** The maximum absolute latitude of the species' geographical range, serving as a reliable proxy for the environmental temperature gradient.
- **Trophic Level:** A numerical value representing the species' position in the marine food web, indicating its dietary and predatory behaviour.

These characteristics were selected because evolutionary biology suggests that brain size in marine teleosts is influenced by baseline allometry (body size) as well as the severe energetic and metabolic constraints imposed by deep-water habitats, temperature, and predatory lifestyles.

The entire dataset is provided in section 7 (Appendix A).

2.1 Summary Statistics

The table below represents the summary statistics for each variable in the dataset.

Variable	Mean	Median	Minimum	Maximum	Std. Dev.
Brain Mass (g)	0.40009	0.258	0.003	3.476	0.46739
Body Mass (g)	437.05290	90.670	0.100	7400.000	967.55306
Maximum Depth (m)	313.13292	60.000	1.000	5998.000	739.82298
Maximum Absolute Latitude	42.37832	35.000	15.000	87.000	15.52173
Trophic Level	3.39489	3.400	2.000	4.500	0.63219

Table 2.1: Summary statistics of each variable

3 Statistical Modelling and Methodology

To investigate the environmental constraints on teleost brain size, a Multiple Linear Regression (MLR) framework is employed. The primary objective is to estimate the simultaneous effects of depth, latitude (temperature proxy), and trophic level on brain mass, while strictly controlling for the baseline allometric scaling of the organism.

3.1 Variable Transformation and Allometry

In biological scaling, the relationship between organ size and body size is rarely linear; it typically follows a power law of the form $Y = cX^k$. To linearize this relationship and stabilize the variance—a necessary condition for Ordinary Least Squares (OLS)—both the response variable (Brain Mass) and the primary covariate (Body Mass) are log-transformed using the natural logarithm.

The inclusion of ln(Body Mass) as an independent variable allows the model to dynamically account for the expected allometric scaling exponent (the β_1 coefficient), effectively eliminating the need for a pre-calculated Encephalization Quotient (EQ) and avoiding mathematical circularity.

3.2 The Multiple Linear Regression Model

The proposed theoretical model for the i-th species is specified as follows:

$$\log(\texttt{BrainMass}_i) = \beta_0 + \beta_1 \log(\texttt{BodyMass}_i) + \beta_2\texttt{Depth}_i + \beta_3\texttt{Latitude}_i + \beta_4\texttt{Trophic}_i + \epsilon_i$$

Where:

- β_0 is the intercept term.
- β_1 represents the allometric scaling exponent.
- $\beta_2, \beta_3, \beta_4$ represent the partial effects of Maximum Depth, Maximum Absolute Latitude, and Trophic Level, respectively, holding all other variables constant.
- ϵ_i is the random error term.

3.3 Model Assumptions and Diagnostics

The validity of the OLS estimates relies on several classical assumptions, which will be rigorously evaluated post-estimation:

1. **Linearity and Additivity:** Assessed by plotting the residuals against the fitted values ($\hat{Y}$) and each independent variable.
2. **Homoscedasticity:** The assumption of constant variance of errors ($\text{Var}(\epsilon_i) = \sigma^2$) will be checked using residual vs. fitted plots and formally tested if necessary. The log-transformation of mass variables inherently mitigates severe heteroscedasticity.

3. **Normality of Errors:** The assumption that $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ will be verified using residual plot and histogram of the residuals.

4. **Multicollinearity:** In marine environments, variables such as depth and latitude may exhibit spatial correlation. To ensure this does not inflate the variance of our coefficient estimates, the pairwise correlations among all explanatory predictors will be calculated and reviewed. The detection of strong intercorrelations between variables will warrant further investigation or potential dimensionality reduction to guarantee the stability and reliability of the final model.

These assumptions were verified using diagnostic plots in residual analysis section (discussed later).

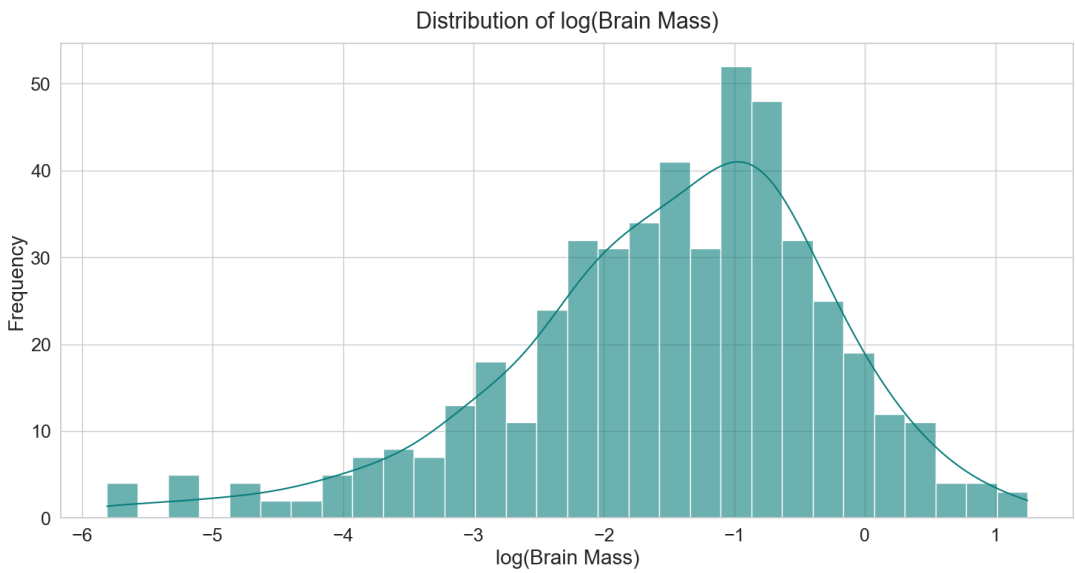

Figure 3.1: Frequency distribution of brain mass

4 Results and Analysis

4.1 Model Estimation

The Multiple Linear Regression (MLR) model was fitted to the data using Ordinary Least Squares (OLS) estimation. The fitted regression equation, utilizing the log-transformed mass variables to account for allometric scaling, is represented as:

$$\log(\widehat{\texttt{BrainMass}}) = -3.5768 + 0.5977\ \log(\texttt{BodyMass}) - 0.0002\ \texttt{Depth} - 0.0067\ \texttt{Latitude} - 0.0831\ \texttt{Trophic}$$

4.2 Regression Output

The results of the regression analysis, including the estimated coefficients, standard errors, test statistics, and corresponding p-values, are summarized in Table 4.1. The model explains approximately 84.4% of the variance in the log-transformed brain mass of the 489 teleost species sampled (Adjusted $R^2 = 0.843$, $p < 0.001$).

Predictor Variable	Coefficient ($\hat{\beta}$)	Std. Error	p-value
Intercept ($\hat{\beta}_0$)	-3.5768	0.132	< 0.001
log(`BodyMass`) ($\hat{\beta}_1$)	0.5977	0.012	< 0.001
`Depth` ($\hat{\beta}_2$)	-0.0002	0.00003	< 0.001
`Latitude` ($\hat{\beta}_3$)	-0.0067	0.002	< 0.001
`Trophic` ($\hat{\beta}_4$)	-0.0831	0.038	0.028

Table 4.1: OLS Regression Results for Teleost Encephalization

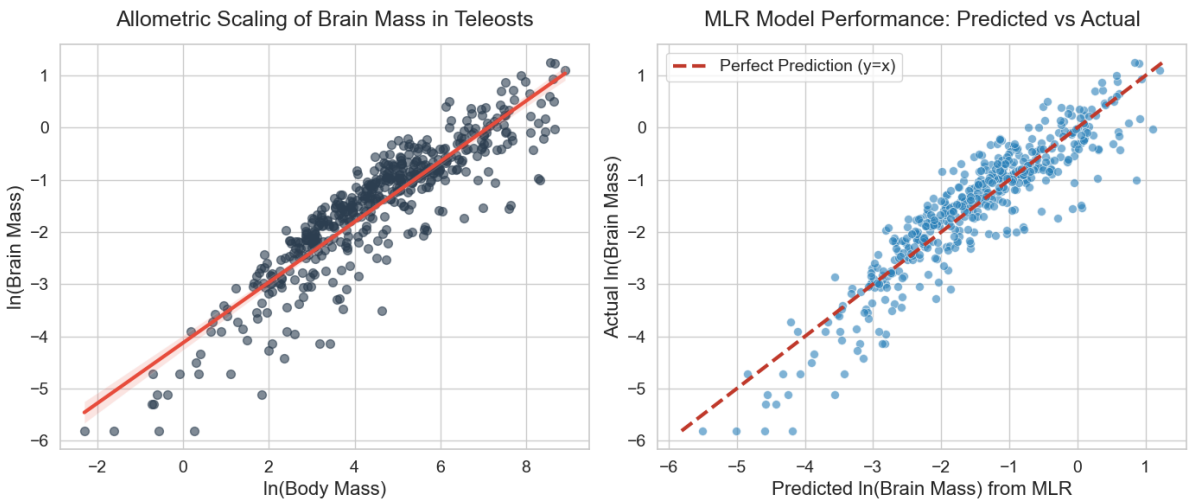

Figure 4.1: Plot showing best fitting line for the Model

Consistent with established principles of evolutionary biology, the allometric scaling of body mass acts as the overwhelming primary driver of encephalization. A simple baseline regression of log-transformed brain mass on log-transformed body mass alone yields an R^2 of 0.809, indicating that physical size independently accounts for nearly 81% of the observed variance. Therefore, the environmental predictors in the full model ($R^2 = 0.844$) should be interpreted as significant but secondary ecological pressures that account for an additional 3.5% of the variance, incrementally modifying the baseline allometric trajectory.

The correlation analysis confirms that body mass is the primary driver of brain size ($r = 0.900$), while environmental factors show much weaker direct associations. No-

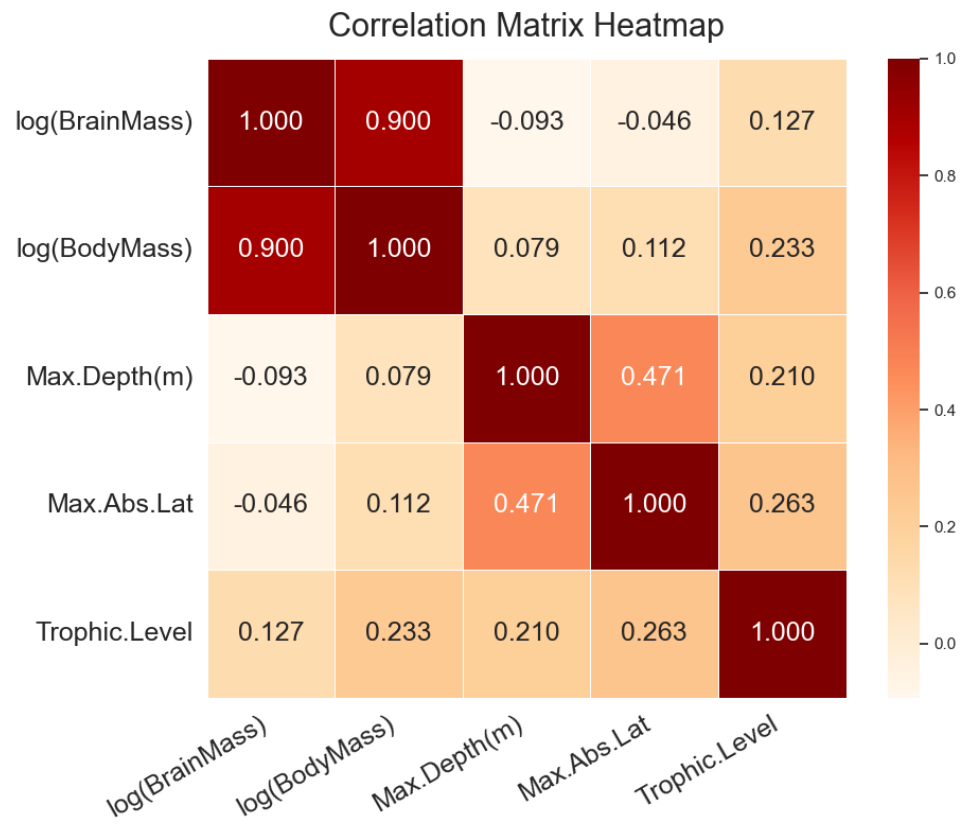

Figure 4.2: Correlation matrix (heatmap).

tably, Maximum Depth and Absolute Latitude are moderately correlated with each other ($r = 0.471$), likely reflecting the shared physical constraints of high-latitude and deep-sea environments. The relatively low correlations between the environmental predictors ($|r| < 0.5$) suggest that multicollinearity is not a significant concern, allowing each ecological variable to contribute independent information to the overall model.

4.3 Interpretation of Findings

The regression results provide highly significant insights into the environmental and ecological constraints acting upon teleost brain allometry. Notably, every predictor variable proved to be statistically significant at the $\alpha = 0.05$ level:

- **Baseline Allometry ($\hat{\beta}_1$):** As expected, body mass is a highly significant predictor of brain mass ($p < 0.001$). The scaling exponent of 0.5977 indicates that for every 1% increase in body mass, brain mass increases by approximately 0.60%, holding all environmental factors constant.

- **The Depth Penalty ($\hat{\beta}_2$):** Habitat depth exhibits a negative and highly significant relationship with relative brain size ($p < 0.001$). This supports the expensive-tissue

hypothesis, confirming that the extreme hydrostatic pressure and resource scarcity of the deep ocean place a strict metabolic tax on the development of energetically costly neural tissue.

- **Temperature Constraints ($\hat{\beta}_3$):** Maximum absolute latitude acts as a proxy for oceanic temperature. The negative coefficient ($p < 0.001$) implies that species inhabiting higher latitudes (colder waters) tend to exhibit smaller relative brain sizes. This is consistent with the biological principle that depressed ambient temperatures lower overall basal metabolic rates, restricting the energy available for brain maintenance.

- **Ecological Paradox of Predation ($\hat{\beta}_4$):** Trophic level is a significant predictor of encephalization ($p = 0.028$). Interestingly, the negative coefficient (-0.0831) indicates that species occupying higher positions in the food web actually exhibit *smaller* brains relative to their body size. This contrasts with some shallow-water evolutionary paradigms and highlights the severe energetic restrictions of the marine environment: the metabolic cost of maintaining a large brain may simply outweigh the cognitive benefits of complex predatory behaviour in these habitats.

5 Residual Analysis and Model Validation

To evaluate the reliability of the OLS estimates and ensure the validity of the associated hypothesis tests, a detailed diagnostic analysis of the model residuals was performed.

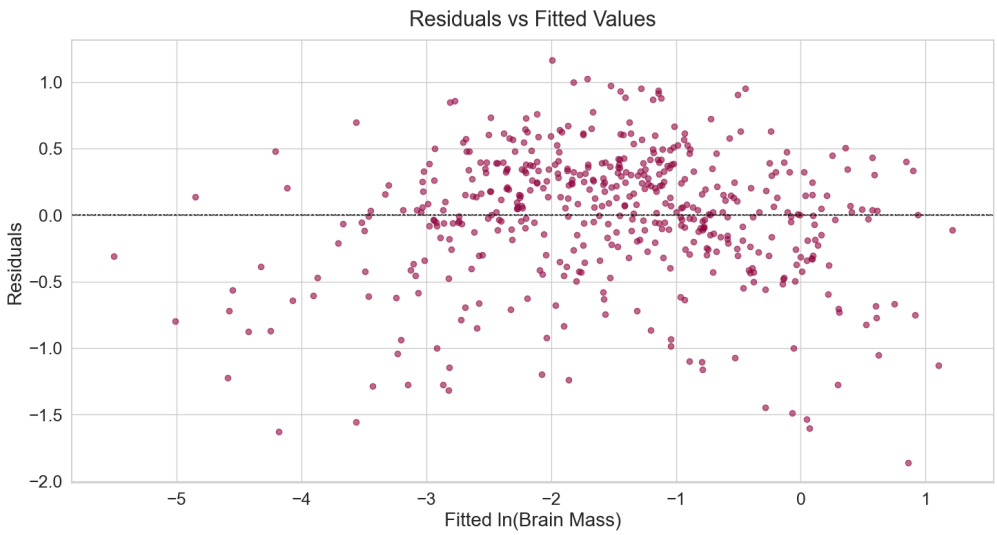

Figure 5.1: Residual plots.

The scatter plot of residuals against fitted values (Figure 5.1) exhibits a random distribution of points centered around the horizontal zero-axis. The absence of any discernible

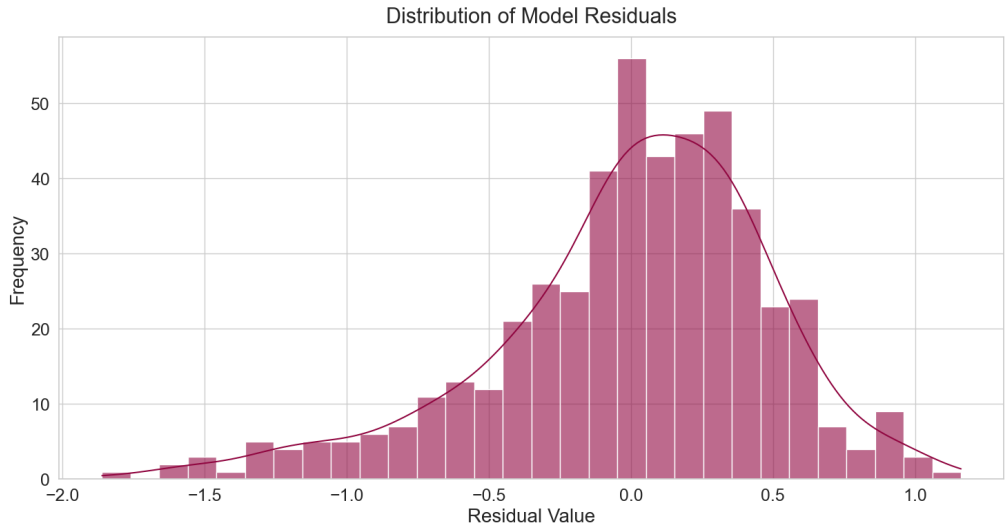

Figure 5.2: Histrogram of residuals.

"U-shaped" or systematic pattern confirms that the relationship between encephalization and the ecological predictors is appropriately captured by the linear functional form. Furthermore, the spread of residuals remains relatively consistent across the range of fitted values, suggesting that the assumption of homoscedasticity is largely satisfied.

Furthermore, the histogram of these residuals (Figure 5.2) provides visual confirmation of the model's consistency. The distribution is centered at zero and exhibits a predominantly bell-shaped profile, which is the hallmark of a well-fitted statistical model. While there is a slight "tail" to the left—representing a small number of species with unexpectedly small brains—the overall symmetry of the graph suggests that the model's errors are balanced. This allows us to conclude that the regression provides a dependable and accurate representation of the factors influencing teleost brain evolution.

6 Conclusion

6.1 Discussion

The primary objective of this analysis was to determine the underlying biological and environmental factors that govern encephalization across diverse teleost species. The multiple linear regression analysis yielded several key insights into the evolutionary pressures shaping marine neuroanatomy.

First and foremost, the analysis confirmed that the fundamental biological law of allometry is the overwhelming primary driver of brain size. The strong positive relationship between log-transformed body mass and brain mass dictates the baseline necessity for

neural tissue, explaining the vast majority of the observed variance independently of external factors.

Beyond physical size, the model highlighted the severe metabolic constraints imposed by the marine environment. The significant negative impacts of both maximum habitat depth and absolute latitude provide strong empirical support for the expensive-tissue hypothesis. The extreme hydrostatic pressures of the deep ocean and the depressed basal metabolic rates associated with colder, high-latitude waters act as strict evolutionary taxes, severely limiting the energy available for the development and maintenance of large, metabolically demanding brains.

Furthermore, the model revealed an intriguing ecological dynamic regarding predatory behavior. The negative coefficient for trophic level indicates that species occupying higher positions in the food web exhibit relatively smaller brains. This suggests that the intense energetic costs of the marine environment may ultimately outweigh the potential cognitive advantages of complex predatory strategies, forcing higher-order predators to prioritize raw physical efficiency over neural complexity.

6.2 Significance of Ecological Predictors

To formally validate the impact of the environmental variables on encephalization, hypothesis testing was conducted for each parameter in the multiple linear regression model. For every predictor, we established a null hypothesis (H_0) positing that the specific variable has no true mathematical effect on brain mass (i.e., the true coefficient is zero).

The p-value associated with each variable quantifies the probability of observing our specific dataset if this null hypothesis were actually true. In biological and ecological modeling, a standard significance threshold (alpha level) of 0.05 is widely adopted.

Upon analyzing the model output, the p-values for all independent variables—log transformed body mass, maximum depth, absolute latitude, and trophic level—were found to be strictly less than the 0.05 threshold ($p < 0.05$). Consequently, we confidently reject the null hypothesis for each of these factors.

This statistical significance proves that the negative coefficients observed for the environmental variables are not artifacts of random sampling noise. Instead, it provides robust empirical evidence that the extreme hydrostatic pressures of deep habitats, the depressed temperatures of high latitudes, and the energetic demands of higher trophic levels exert genuine, measurable constraints on the evolution of teleost brain size.

6.3 Biological Interpretation: The Encephalization Quotient

In evolutionary biology, absolute brain mass is rarely used as a standalone metric for cognitive capacity, as it is heavily confounded by the sheer physical size of the organism. Instead, biologists utilize the Encephalization Quotient (EQ), a measure that quantifies how much larger or smaller a species' brain is compared to the expected baseline for its body mass.

Our linear regression model provides a direct, mathematical framework for calculating and interpreting this biological concept. The foundational allometric relationship between brain and body mass is expressed as Brain Mass $= c \times (\text{Body Mass})^a$. By applying a logarithmic transformation, this biological law translates precisely into the primary linear component of our model: $\ln(\text{Brain Mass}) = \ln(c) + a \cdot \ln(\text{Body Mass})$.

Consequently, the residuals extracted from our regression analysis hold profound biological significance. A positive residual indicates that a teleost species possesses a higher encephalization quotient than average (a larger brain than predicted by its size), while a negative residual indicates a lower EQ.

By incorporating environmental predictors into the model, we are fundamentally analyzing the variance in EQ. The significant negative coefficients for maximum depth, absolute latitude, and trophic level demonstrate that these specific ecological pressures are responsible for driving the Encephalization Quotient downward. Ultimately, this statistical framework proves that while body mass dictates the baseline expectation for neural tissue, the harsh energetic constraints of a species' specific marine habitat dictate its final EQ.

6.4 Summary

In summary, this project successfully developed a robust predictive model for teleost brain mass by integrating allometric scaling with key environmental constraints. Rigorous visual diagnostic checks confirmed that the linear model satisfied all core statistical assumptions regarding normality and multicollinearity, ensuring the reliability of the estimated parameters. The overarching conclusion is that while body size provides the foundational architectural blueprint for encephalization, the harsh and energetically restrictive realities of the marine environment—specifically depth, temperature, and trophic position—act as the ultimate evolutionary sculptors of teleost neuroanatomy.

7 Appendix A: Data for Teleost Encephalization Analysis

This appendix contains the raw dataset used for the multiple linear regression analysis, including the body mass, brain mass, and associated environmental variables (maximum depth, absolute latitude, and trophic level) for the teleost species studied.

Genus	species	Max.Depth(m)	Max.Abs.Lat	Trophic.Level	Body.Mass(g)	Brain.Mass(g)
Ablennes	hians	3	38	4.5	1500	1.018
Abudefduf	abdominalis	50	29	3	75.2	0.234
Abudefduf	saxatilis	20	41	3.4	89.53	0.356
Abudefduf	septemfasciatus	3	35	3	200	0.304
Abudefduf	sexfasciatus	20	35	2.4	41.38	0.182
Abudefduf	sordidus	3	37	2.8	200	0.4
Abudefduf	vaigiensis	15	39	2.6	13.8	0.099
Abudefduf	whitleyi	5	33	2.6	165	0.377
Acanthostracion	polygonius	80	41	2	320	0.466
Acanthurus	bahianus	40	43	2	107	0.405
Acanthurus	chirurgus	25	37	2	35	0.218
Acanthurus	coeruleus	40	42	2	118	0.49
Acanthurus	dussumieri	131	40	2	451.23	0.673
Acanthurus	guttatus	6	30	2	288	0.662
Acanthurus	mata	100	35	2.5	255.85	0.484
Acanthurus	nigricans	67	30	2	264	0.548
Acanthurus	nigrofuscus	25	34	2	48.93	0.218
Acanthurus	nigroris	90	28	2	98.7	0.373
Acanthurus	pyroferus	60	32	2	154	0.404
Acanthurus	thompsoni	119	29	3.4	148.75	0.368
Acanthurus	triostegus	90	36	2.8	64.18	0.271
Acanthurus	xanthopterus	100	30	2.9	677.25	0.537
Acipenser	transmontanus	122	58	3.2	3500	0.65
Aeoliscus	strigatus	20	32	3.5	1.9	0.02
Agonus	cataphractus	270	77	3.4	8	0.043
Alosa	alosa	300	61	3.6	725	0.43
Alosa	fallax	300	70	3.6	600	0.452
Alosa	immaculata	90	50	3.9	335	0.421
Aluterus	scriptus	120	46	2.8	158	0.41
Amblyglyphidodon	curacao	40	30	2.6	21.53	0.138
Amphiprion	allardi	30	30	2.7	32.2	0.102
Amphiprion	bicinctus	30	31	2.7	21.23	0.06
Amphiprion	chrysopterus	30	15	2.8	22	0.081
Amphiprion	melanopus	18	30	2.8	27.07	0.096
Anampses	twistii	30	30	3.5	20.6	0.109
Anisotremus	surinamensis	20	33	3.3	1067	1.274
Anoplogaster	cornuta	4992	65	4	114.5	0.08
Anyperodon	leucogrammicus	80	32	3.9	375	0.476
Aphareus	furca	122	35	4.1	127.5	0.719
Apogon	exostigma	40	30	3.7	14.85	0.084
Apogon	kallopterus	158	32	3.5	22.8	0.123
Apogon	maculatus	128	43	3.5	9.8	0.05
Aprion	virescens	180	35	4	1665	2.018
Archamia	fucata	60	30	3.5	5.7	0.049
Argentina	silus	1440	75	3.3	279.5	0.493
Argentina	sphyraena	700	70	3.6	67.33	0.204
Argyropelecus	aculeatus	600	40	3.5	16.5	0.039
Argyrops	spinifer	150	34	4.5	1800	1.267
Aristostomias	scintillans	1219	66	3.5	10.5	0.012
Arothron	hispidus	50	35	3.1	194.07	0.209
Arothron	immaculatus	17	37	2.5	66.5	0.136
Arothron	nigropunctatus	25	42	3.3	213	0.359
Arothron	stellatus	58	32	3.3	5775	0.969
Aspidophoroides	monopterygius	695	84	3	20.67	0.062
Aspitrigla	cuculus	400	60	3.9	627.5	0.749
Asterropteryx	semipunctata	20	32	2.3	0.48	0.005
Atherinomorus	lacunosus	39	32	3.3	21.55	0.102
Atule	mate	80	35	4.5	160.37	0.567

Aulostomus	chinensis	122	32	4.2	217.64	0.411
Balistapus	undulatus	50	32	3.4	470	0.527
Balistes	capriscus	100	58	3.6	873.9	0.67
Balistoides	conspicillum	75	43	3.3	872.5	0.917
Bathygobius	cocosensis	10	32	3.5	1.35	0.011
Beryx	decadactylus	1000	70	4.1	691.67	0.806
Beryx	splendens	1300	45	4.4	257.5	0.728
Boops	boops	350	63	3	122	0.433
Boreogadus	saida	400	87	3.1	47.5	0.209
Brama	brama	1000	70	4.1	1850	1.261
Brosme	brosme	1000	83	4	583.1	0.588
Brotula	barbata	650	30	3.8	1.2	0.02
Brotula	multibarbata	650	38	3.7	344.3	0.418
Bryaninops	natans	27	30	3	0.7	0.006
Caesio	caerulaurea	50	34	3.4	290	0.698
Caesio	cuning	60	31	3.4	335	0.611
Callionymus	lyra	430	65	3.3	18.26	0.048
Callionymus	maculatus	650	65	3.3	13.5	0.054
Callogobius	sclateri	35	32	3.2	8	0.016
Calotomus	spinidens	10	30	2	157	0.307
Cantherhines	dumerilii	70	32	3.1	380.5	0.467
Cantherhines	pullus	50	43	2.6	52	0.242
Canthigaster	jactator	30	30	2.9	17.5	0.082
Canthigaster	rostrata	40	30	3	3.4	0.046
Canthigaster	solandri	36	30	3	8.03	0.058
Canthigaster	valentini	55	32	2.8	6.45	0.051
Capros	aper	700	62	3.1	29	0.172
Caracanthus	maculatus	15	35	3.2	2	0.024
Carangoides	bartholomaei	50	42	4.5	57.6	0.44
Caranx	ignobilis	188	37	4.2	3200	1.906
Caranx	melampygus	190	35	4.5	2185.19	2.046
Caranx	sexfasciatus	146	36	4.5	520	1.155
Careproctus	reinhardti	1750	81	3.8	90.67	0.082
Centroberyx	gerrardi	500	40	4.1	663.05	0.819
Centrolophus	niger	1050	68	3.9	955	0.363
Centropogon	australis	30	38	3.2	6.2	0.035
Centropyge	bicolor	25	35	3	17.5	0.116
Centropyge	bispinosa	45	35	2.8	22.1	0.108
Centropyge	flavissima	50	35	2.8	29.8	0.151
Centropyge	potteri	138	30	2.6	45.3	0.14
Centropyge	tibicen	55	32	2.8	12.5	0.11
Cephalopholis	argus	40	34	4.5	1040	0.667
Cephalopholis	boenak	64	32	4.2	82.33	0.176
Cephalopholis	cruentata	170	36	4.2	195.2	0.4
Cephalopholis	fulva	150	36	4.1	156.53	0.381
Cephalopholis	miniata	150	35	4.4	324	0.368
Cephalopholis	urodeta	60	34	4	97.5	0.262
Cepola	macrophthalma	400	61	3.1	25	0.087
Cetoscarus	bicolor	30	33	2	1120	0.86
Chaetodon	auriga	35	30	3.2	132.7	0.403
Chaetodon	baronessa	20	45	3.3	79	0.2
Chaetodon	bennetti	30	28	3.3	68.7	0.27
Chaetodon	capistratus	20	37	3	22.5	0.188
Chaetodon	citrinellus	36	32	3.1	42	0.223
Chaetodon	ephippium	30	30	3.1	124.88	0.377
Chaetodon	flavirostris	20	30	3	73.35	0.353
Chaetodon	kleinii	61	30	3.1	29.07	0.169
Chaetodon	lunula	30	32	3.4	137.75	0.478
Chaetodon	melannotus	20	36	3.3	20	0.176
Chaetodon	mertensii	120	33	3	24.1	0.166
Chaetodon	ocellatus	30	37	3.2	88.53	0.372
Chaetodon	ornatissimus	36	30	3.3	84.35	0.313
Chaetodon	plebeius	10	45	3.3	17.6	0.129
Chaetodon	quadrimaculatus	15	30	3.3	51.2	0.261
Chaetodon	reticulatus	30	30	2.6	81.17	0.292
Chaetodon	speculum	30	30	3.6	18	0.133
Chaetodon	striatus	55	37	3.2	98	0.347
Chaetodon	trifascialis	30	30	3.3	77	0.333
Chaetodon	trifasciatus	20	30	3.3	41.17	0.208
Chaetodon	ulietensis	30	32	3.3	35.5	0.231
Chaetodon	unimaculatus	60	35	3.1	23.1	0.152
Channichthys	rhinoceratus	750	54	3.3	375	0.39

Chauliodus	sloani	4700	70	4.2	41.33	0.031
Cheilinus	chlorourus	30	30	3.4	127.15	0.252
Cheilinus	fasciatus	60	30	3.4	246.6	0.416
Cheilinus	trilobatus	30	30	3.5	270	0.416
Cheilio	inermis	30	30	4	126.7	0.372
Cheilodipterus	macrodon	40	30	4	21.93	0.108
Chelidonichthys	lucerna	318	66	3.7	528.8	0.447
Chlorurus	sordidus	50	33	2	332.83	0.445
Chromis	agilis	65	31	3	9.7	0.079
Chromis	amboinensis	70	23	2.7	13.2	0.079
Chromis	atripectoralis	29	32	3.1	21.1	0.113
Chromis	atripes	40	35	2.6	6.8	0.047
Chromis	chromis	40	46	3	20	0.135
Chromis	chrysura	45	35	3	20.3	0.11
Chromis	cyanea	60	32	3.1	6.64	0.088
Chromis	flavomaculata	40	35	3	35	0.215
Chromis	multilineata	60	30	3	25	0.18
Chromis	retrofasciata	65	30	2.7	2.6	0.033
Chromis	ternatensis	36	35	3.4	13.85	0.113
Chromis	weberi	40	32	3.4	20	0.163
Chrysiptera	brownriggii	12	32	2.7	5.2	0.051
Chrysiptera	glauca	3	30	2.4	7.9	0.061
Chrysiptera	rollandi	35	23	2.7	2.1	0.028
Ciliata	mustela	22	72	3.5	9.84	0.055
Ciliata	septentrionalis	90	70	3.5	10	0.047
Cirrhitus	pinnulatus	23	30	3.6	161	0.237
Clupea	harengus	364	80	3.2	154.83	0.319
Coelorinchus	caelorhincus	1250	62	3.6	120.65	0.359
Coris	aygula	30	30	3.4	1510	0.656
Coris	flavovittata	98	30	3.2	17.9	0.117
Coris	gaimard	50	37	3.5	42.2	0.16
Coris	julis	120	59	3.2	14.64	0.113
Coryphaena	hippurus	85	47	4.4	5740	3.423
Coryphaenoides	rupestris	2600	66	3.5	721.89	0.651
Corythoichthys	intestinalis	68	23	3.2	2.4	0.02
Ctenochaetus	binotatus	53	34	2	48.5	0.245
Ctenochaetus	striatus	35	35	2	94.8	0.272
Ctenochaetus	strigosus	113	30	2	52.97	0.218
Ctenolabrus	rupestris	50	64	3.4	15.06	0.111
Cyclopterus	lumpus	868	80	3.9	1982.5	0.211
Dactyloptena	orientalis	100	37	3.7	20.93	0.173
Dactylopterus	volitans	100	52	3.6	11	0.14
Dascyllus	albisella	50	30	3.1	47.65	0.166
Dascyllus	aruanus	20	38	2.5	5.07	0.048
Dascyllus	reticulatus	50	30	3.1	10.7	0.083
Dascyllus	trimaculatus	55	30	2.8	70.7	0.201
Decapterus	macrosoma	214	32	3.4	150	0.785
Decapterus	tabl	400	37	3.2	460	1.492
Dendrochirus	brachypterus	68	32	3.6	40	0.065
Dicentrarchus	labrax	100	72	3.8	449	0.367
Diodon	holocanthus	200	39	3.3	593	0.731
Diodon	hystrix	50	35	3.4	835.75	0.878
Diplodus	annularis	90	49	3.4	82.67	0.31
Dissostichus	eleginoides	3850	66	4	3278.4	0.907
Doryrhamphus	dactyliophorus	56	33	3.8	0.1	0.003
Echeneis	naucrates	50	45	3.4	1282.5	1.039
Echidna	nebulosa	48	32	4	44.7	0.045
Enchelyopus	cimbrius	650	73	3.5	50	0.169
Epibulus	insidiator	42	30	3.8	148.9	0.297
Epigonus	telescopus	1200	65	3.3	111.04	0.503
Epinephelus	corallicola	30	30	3.8	226	0.41
Epinephelus	cyanopodus	150	36	4.1	587.5	0.447
Epinephelus	fasciatus	160	36	3.7	157	0.302
Epinephelus	hexagonatus	30	33	4.1	126.5	0.256
Epinephelus	longispinis	70	33	4.2	65	0.247
Epinephelus	maculatus	100	34	3.9	953.33	0.581
Epinephelus	malabaricus	150	32	3.8	44	0.104
Epinephelus	melanostigma	30	34	4	274	0.406
Epinephelus	merra	50	35	3.8	62.92	0.181
Epinephelus	polyphekadion	46	34	3.9	1676	0.667
Epinephelus	tauvina	300	32	4.1	1205	0.745
Etelis	oculatus	450	36	4.2	1750	1.425

Eucinostomus	gula	55	51	3.2	17	0.15
Eutrigla	gurnardus	340	70	3.6	444.71	0.564
Eviota	prasina	2	32	3.1	0.2	0.003
Exyrias	puntang	5	35	3.5	19.55	0.048
Fistularia	commersonii	132	32	4.3	217.78	0.489
Fistularia	petimba	200	44	4.5	500	1.005
Forcipiger	flavissimus	145	36	3.1	39.17	0.276
Forcipiger	longirostris	70	28	3.5	70.4	0.313
Fowleria	aurita	37	35	3.5	2.8	0.027
Gadus	morhua	600	80	4.4	4237.52	1.588
Gadus	ogac	200	79	3.6	1983.3	1.402
Gaidropsarus	argentatus	2260	72	3.4	47.33	0.173
Gaidropsarus	ensis	2000	76	3.4	71.33	0.152
Gaidropsarus	mediterraneus	450	62	3.4	458.1	0.508
Gerres	cinereus	15	32	3.3	133	0.482
Gerres	filamentosus	50	32	3.3	40	0.215
Glaucosoma	scapulare	90	35	4.3	1100	0.73
Gnathodentex	aureolineatus	30	35	3.3	172	0.707
Gobionotothen	acuta	300	54	3.3	109	0.182
Gobius	cobitis	35	53	3	46.1	0.095
Gobius	niger	75	66	3.2	14.8	0.034
Gobius	paganellus	15	60	3.3	7.08	0.029
Gomphosus	varius	30	30	3.6	50.35	0.225
Grammistes	sexlineatus	130	32	4	75	0.099
Gymnocranius	grandoculis	170	35	3.4	1044	1.42
Gymnocranius	griseus	80	35	3.2	58	0.409
Gymnothorax	buroensis	25	30	3.8	106	0.131
Gymnothorax	flavimarginatus	150	36	4.2	693	0.177
Gymnothorax	javanicus	50	30	3.9	4090	0.367
Gymnothorax	margaritophorus	25	30	3.6	207	0.136
Gymnothorax	melatremus	58	28	3.7	3	0.009
Gymnothorax	meleagris	51	38	4.5	140.5	0.383
Gymnothorax	pictus	100	30	4.2	425	0.202
Gymnothorax	rueppellii	40	30	4	32.6	0.052
Gymnothorax	undulatus	110	32	3.6	1094	0.216
Gymnothorax	zonipectis	40	19	3.9	11.2	0.032
Haemulon	carbonarium	25	33	3.3	192	0.694
Haemulon	flavolineatum	60	34	3.3	96	0.474
Haemulon	plumierii	40	39	3.6	55.7	0.32
Halargyreus	johnsonii	3000	63	3.4	33.87	0.147
Halichoeres	biocellatus	35	32	3.4	16.15	0.086
Halichoeres	bivittatus	15	37	3.3	18.63	0.117
Halichoeres	garnoti	80	32	3.5	38	0.167
Halichoeres	hortulanus	30	32	3.4	130.87	0.294
Halichoeres	maculipinna	24	37	3.3	12	0.064
Halichoeres	margaritaceus	5	32	3.7	6.1	0.068
Halichoeres	marginatus	30	31	3.3	18.5	0.088
Halichoeres	poeyi	15	33	3.4	26	0.112
Halichoeres	trimaculatus	30	32	3.5	5.87	0.062
Hemigymnus	fasciatus	20	30	3.2	231	0.528
Hemigymnus	melapterus	30	32	3.3	196	0.432
Hemitaurichthys	polylepis	40	32	3.1	34.3	0.19
Heniochus	acuminatus	75	35	3.5	69.25	0.258
Heniochus	chrysostomus	40	32	3.3	78.5	0.318
Heniochus	monoceros	30	32	3.5	272	0.481
Heniochus	singularius	40	32	3.6	248	0.406
Heniochus	varius	20	32	3.2	95	0.308
Heteropriacanthus	cruentatus	300	33	3.8	226.67	0.518
Hyperoplus	lanceolatus	30	80	4.2	16.3	0.047
Hyporthodus	septemfasciatus	30	40	4	8	0.051
Idiacanthus	fasciola	2000	54	3.9	6.3	0.006
Knipowitschia	caucasica	2	47	3.3	0.5	0.005
Kuhlia	mugil	18	32	3.8	45	0.203
Kyphosus	cinerascens	45	35	2.3	1625	1.539
Labrichthys	unilineatus	20	21	3.3	20	0.1
Labroides	dimidiatus	40	30	3.5	7.58	0.059
Labrus	bergylta	50	68	3.1	366.71	0.448
Labrus	mixtus	200	71	3.9	136.9	0.285
Lactophrys	bicaudalis	50	33	3	114	0.386
Lepidonotothen	mizops	220	53	3.2	33	0.151
Lepidonotothen	squamifrons	900	56	3.1	262.5	0.311
Lepidotrigla	papilio	60	43	3.5	72	0.142

Lesueurigobius	suerii	337	46	3.6	0.57	0.003
Lethrinus	genivittatus	25	35	3.5	700	0.943
Lethrinus	lentjan	90	32	4.2	500	1.663
Lethrinus	microdon	80	32	3.8	2250	1.704
Lethrinus	obsoletus	30	28	3.4	610	0.856
Lethrinus	xanthochilus	150	33	3.7	1840	2.367
Lipophrys	pholis	8	65	3.1	9.33	0.051
Lophius	budegassa	1013	59	4.5	270	0.127
Lophius	piscatorius	1000	75	4.5	3917.33	0.375
Lutjanus	apodus	63	38	4.2	87	0.576
Lutjanus	argentimaculatus	120	32	3.6	7400	2.992
Lutjanus	bohar	180	33	4.1	2895	2.439
Lutjanus	buccanella	200	42	3.9	34	0.258
Lutjanus	fulvus	75	34	4.1	161.44	0.495
Lutjanus	gibbus	150	33	3.6	825	1.437
Lutjanus	griseus	180	41	4.3	233	0.632
Lutjanus	kasmira	265	35	3.6	240	0.675
Lutjanus	mahogoni	100	36	4.4	108	0.594
Lutjanus	monostigma	60	31	4.3	831	1.3
Lutjanus	rivulatus	100	35	4.1	190	0.52
Lutjanus	vitta	72	35	4.1	107	0.507
Macolor	niger	90	31	4	2620	2.724
Macrourus	berglax	1000	82	4.5	774.22	0.998
Malacocephalus	laevis	1000	62	4.2	210.72	0.447
Malacosteus	niger	3886	66	3.7	24	0.016
Mallotus	villosus	725	79	3.2	23.33	0.094
Melanogrammus	aeglefinus	450	79	4.1	1413.78	1.131
Melichthys	vidua	60	32	3.4	81.5	0.341
Mene	maculata	200	32	3.5	24.8	0.162
Merlangius	merlangus	200	72	4.4	202.92	0.229
Micromesistius	poutassou	3000	79	4	162	0.435
Microspathodon	chrysurus	120	32	2.1	53	0.198
Mola	mola	480	65	4	4600	0.922
Molva	dypterygia	1000	79	4.5	3311.85	1.238
Molva	macrophthalma	754	61	4.5	2173.33	0.785
Molva	molva	1000	75	4.3	5724.68	1.659
Monotaxis	grandoculis	100	35	3.2	115.08	0.418
Mora	moro	2500	64	3.8	1124.63	0.628
Mullus	surmuletus	409	62	3.4	228.33	0.539
Muraenolepis	marmorata	1600	56	3.4	109.9	0.323
Myctophum	punctatum	1000	69	3.4	7.37	0.042
Myoxocephalus	scorpius	451	80	3.9	386.25	0.206
Myripristis	berndti	159	32	3.7	24.9	0.209
Myripristis	jacobus	100	37	3.6	119	0.384
Myripristis	kuntee	55	32	3.3	27.27	0.22
Myripristis	violacea	30	30	3.5	88	0.341
Nemichthys	scolopaceus	3656	55	3.5	36	0.037
Nemipterus	furcosus	110	34	3.6	218	0.684
Neoglyphidodon	nigroris	23	30	3	47	0.124
Neoniphon	aurolineatus	160	32	4	190.65	0.602
Neoniphon	sammara	46	30	3.6	86.4	0.441
Neopomacentrus	cyanomos	30	35	3.4	15	0.1
Nerophis	lumbriciformis	30	63	4	1.3	0.003
Nezumia	aequalis	2320	70	3.3	33.33	0.121
Notacanthus	bonaparte	2000	55	3.4	114	0.143
Notacanthus	chemnitzii	3285	61	3.5	4500	0.812
Notothenia	rossii	350	67	3.6	1880	0.929
Oblada	melanura	30	44	3	325	0.633
Odonus	niger	40	32	3.2	1045	0.786
Omosudis	lowii	4000	40	4.3	60	0.058
Oxycheilinus	digramma	50	30	3.7	88.75	0.26
Oxycheilinus	unifasciatus	160	30	4.1	108.23	0.252
Pagellus	bogaraveo	700	65	3.7	1157.14	1.47
Pagellus	erythrinus	300	60	3.4	123.5	0.598
Parablennius	gattorugine	32	59	2.9	59.53	0.111
Parablennius	sanguinolentus	1	50	2.1	46	0.101
Paracirrhites	arcatus	91	32	3.6	13.3	0.091
Parapercis	clathrata	50	30	3.6	12	0.085
Parapercis	cylindrica	20	30	3.1	14.5	0.089
Parastromateus	niger	105	42	2.9	400	0.666
Pempheris	oualensis	36	30	3.6	78.28	0.26
Pentapodus	caninus	30	33	3.6	165	0.497

Pholis	gunnellus	100	72	3.5	13.45	0.019
Phycis	blennoides	1047	71	3.7	1820.7	0.914
Platax	orbicularis	30	32	3.3	900	0.995
Plectroglyphidodon	dickii	15	30	3.4	12.2	0.114
Plectroglyphidodon	lacrymatus	40	30	2.2	17.5	0.122
Plectroglyphidodon	leucozonus	6	30	2	38.8	0.112
Plectropomus	laevis	100	33	4.1	480	0.573
Plectropomus	leopardus	100	35	4.5	3978.75	1.085
Plectropomus	maculatus	100	28	4.1	4600	1.177
Plotosus	lineatus	60	38	3.5	38.75	0.195
Pollachius	pollachius	200	72	4.2	188.75	0.291
Pollachius	virens	364	77	4.4	5088.5	1.834
Polydactylus	plebeius	122	37	3.6	1210	1.094
Pomacanthus	imperator	100	31	2.7	663.17	0.515
Pomacanthus	paru	100	30	2.8	88	0.23
Pomacanthus	semicirculatus	30	30	2.5	1900	0.739
Pomacanthus	sexstriatus	50	30	2.6	430	0.414
Pomacentrus	pavo	16	32	3	5.5	0.058
Pomacentrus	philippinus	12	30	2.7	23.7	0.154
Pomacentrus	vaiuli	45	33	3.1	6.6	0.069
Pomatoschistus	marmoratus	70	48	3.4	0.55	0.006
Pomatoschistus	microps	12	64	3.3	0.93	0.009
Pomatoschistus	minutus	200	71	3.2	4.46	0.017
Pomatoschistus	pictus	55	65	3.1	1.43	0.009
Priacanthus	hamrur	250	32	3.6	433	0.666
Pseudanthias	squamipinnis	55	32	3.4	9.48	0.064
Pseudobalistes	fuscus	50	32	3.5	1180	0.912
Pseudocoris	yamashiroi	25	32	3.4	2.8	0.031
Pseudogramma	polyacantha	15	32	3.7	5.65	0.026
Pseudupeneus	maculatus	90	40	3.5	167	0.464
Ptereleotris	microlepis	50	30	3.4	4	0.021
Pterocaesio	digramma	50	30	3.4	158	0.575
Pterocaesio	pisang	100	30	3.4	195	0.728
Pterois	antennata	50	42	3.6	282.29	0.15
Pterois	volitans	55	43	4.5	38	0.038
Pungitius	pungitius	110	82	3.3	0.49	0.009
Pygoplites	diacanthus	48	30	2.7	51.4	0.202
Rhabdosargus	sarba	60	38	3.4	58.54	0.223
Rhinecanthus	aculeatus	50	30	3.3	143.68	0.318
Rhinecanthus	rectangulus	20	32	3.1	160.7	0.532
Salarias	fasciatus	8	30	2	9.87	0.056
Salmo	salar	210	72	4.4	280	0.428
Sardina	pilchardus	100	68	3.1	160	0.354
Sardinops	sagax	200	61	2.4	52.31	0.195
Sargocentron	caudimaculatum	40	32	3.9	111.05	0.364
Sargocentron	diadema	60	32	3.4	25.13	0.187
Sargocentron	microstoma	183	35	3.6	40.65	0.275
Sargocentron	punctatissimum	183	32	3.4	37.06	0.24
Sargocentron	spiniferum	122	38	3.5	383.5	0.546
Sargocentron	tiere	183	32	3.5	164.8	0.497
Saurida	undosquamis	350	40	4.5	48.7	0.071
Scarus	frenatus	20	32	2	395	0.455
Scarus	ghobban	36	32	2	2095	0.846
Scarus	globiceps	12	30	2	685	0.739
Scarus	niger	15	30	2	337.5	0.519
Scarus	oviceps	15	30	2	208.5	0.348
Scarus	prasiognathos	15	30	2	1060	0.686
Scarus	psittacus	25	36	2	513.67	0.571
Scarus	rubroviolaceus	36	30	2	754.4	0.66
Scarus	taeniopterus	25	33	2	57.33	0.179
Scatophagus	argus	4	32	3	137	0.242
Schedophilus	medusophagus	358	64	4	1455	0.348
Scolopsis	vosmeri	25	32	3.5	29.5	0.156
Scorpaena	plumieri	60	41	3.8	190	0.132
Scorpaena	porcus	800	55	3.9	117.81	0.11
Scorpaena	scrofa	500	60	4.3	293.75	0.139
Scorpaenopsis	diabolus	70	32	4.2	278.6	0.142
Sebastes	mentella	1441	79	3.7	1750	1.062
Sebastes	viviparus	300	70	4	395	0.447
Selar	crumenophthalmus	170	47	4.1	187	0.816
Selaroides	leptolepis	25	39	3.5	134	0.453
Seriola	rivoliana	160	43	4.5	5200	3.476

Seriolina	nigrofasciata	150	40	4.2	5600	2.554
Serranus	cabrilla	500	57	3.4	21.16	0.12
Serranus	hepatus	100	45	3.5	21.25	0.13
Serranus	scriba	150	50	3.8	29.5	0.115
Serrivomer	beanii	5998	60	3.9	103.5	0.03
Siganus	argenteus	40	30	2	239.17	0.386
Siganus	canaliculatus	50	35	2.8	210	0.347
Siganus	doliatus	5	30	2	110	0.437
Siganus	luridus	40	35	2	391	0.435
Siganus	puellus	30	30	3	75	0.334
Siganus	punctatus	40	30	2.8	169	0.372
Siganus	rivulatus	30	35	2	78.4	0.183
Siganus	spinus	50	30	2	38.6	0.231
Siganus	vulpinus	30	30	2.7	154	0.377
Sigmops	bathyphilus	3000	66	3.5	30.5	0.016
Simenchelys	parasitica	2620	44	3.7	91	0.244
Sparisoma	aurofrenatum	20	33	2	237	0.432
Sparisoma	viride	50	34	2	621	0.544
Sphyraena	barracuda	100	42	4.5	1176	0.98
Sphyraena	forsteri	300	30	4.3	735	1.148
Sphyraena	obtusata	120	32	4.5	37.7	0.187
Spicara	maena	130	47	4.2	72.98	0.267
Spondyliosoma	cantharus	300	63	3.3	1499	1.204
Spratelloides	delicatulus	50	40	3.1	1.5	0.013
Sprattus	sprattus	150	66	3	15.5	0.098
Stegastes	nigricans	12	30	2.2	33.37	0.153
Stenobrachius	leucopsarus	3400	66	3.2	3.53	0.024
Stethojulis	bandanensis	30	32	3.2	26	0.154
Sufflamen	bursa	90	30	3.1	61.1	0.3
Sufflamen	chrysopterum	30	32	3.5	287.67	0.322
Symphodus	mediterraneus	50	45	3.1	22.88	0.113
Symphodus	melanocercus	25	46	3	7	0.075
Symphodus	ocellatus	30	47	3.3	25.5	0.151
Symphodus	rostratus	50	47	3.5	12	0.115
Symphodus	tinca	50	44	3.1	77.24	0.234
Synanceia	verrucosa	30	30	4.2	1100	0.87
Synaphobranchus	kaupii	4800	69	4.1	19.42	0.057
Syngnathus	acus	110	71	3.4	11.17	0.02
Syngnathus	typhle	20	71	4.3	7.27	0.014
Synodus	saurus	400	45	4.5	45	0.111
Synodus	variegatus	121	45	4.2	58.56	0.125
Taenianotus	triacanthus	134	30	4.1	5.5	0.029
Taurulus	bubalis	200	72	3.6	16.13	0.062
Terapon	jarbua	350	32	3.9	295	0.435
Thalassoma	amblycephalum	15	32	3.1	26.2	0.171
Thalassoma	jansenii	15	47	3.3	77	0.265
Thalassoma	lunare	20	32	3.5	21.5	0.12
Thalassoma	lutescens	30	32	3.4	40.7	0.176
Thalassoma	quinquevittatum	40	32	3.6	17.6	0.137
Trachinocephalus	myops	400	40	4.4	92.75	0.115
Trachinus	draco	150	66	4.2	173.9	0.221
Trachurus	novaezelandiae	500	50	3.2	22	0.13
Trachurus	trachurus	1050	67	3.6	292.4	0.798
Trachyrincus	scabrus	1700	55	3.5	304.97	0.321
Trachyscorpia	cristulata	1100	60	3.9	2087.14	0.226
Trigla	lyra	700	60	3.5	320	0.354
Trigloporus	lastoviza	150	70	3.4	257.33	0.378
Trisopterus	esmarkii	300	79	3.2	41.5	0.228
Trisopterus	luscus	100	62	3.7	169.07	0.348
Trisopterus	minutus	440	66	3.8	71.83	0.302
Upeneus	sulphureus	90	40	3.2	18	0.102
Upeneus	tragula	50	30	3.6	43.7	0.203
Valenciennea	strigata	25	32	4	7.6	0.05
Variola	louti	250	37	4.3	297	0.454
Xenodermichthys	copei	2650	70	3.2	27.93	0.064
Zanclorhynchus	spinifer	400	55	3.3	169.5	0.222
Zanclus	cornutus	182	37	2.9	132.2	0.503
Zebrasoma	flavescens	46	30	2	79.25	0.265
Zebrasoma	scopas	60	36	2	57.85	0.248
Zebrasoma	velifer	30	36	2	448.63	0.539
Zeus	faber	400	75	4.5	1217	0.43

8 Appendix B: Code and Computational Framework

All the codes and a CSV version of the dataset are provided here.

9 Reference

[1] Froese, R., and Pauly, D. (Eds.). (2023). *FishBase*. World Wide Web electronic publication. Available at: www.fishbase.org.

[2] Iglésias, T. L., Dornburg, A., Brandley, M. C., Alfaro, M. E., & Warren, D. L. (2015). Life in the unthinking depths: energetic constraints on encephalization in marine fishes. *Journal of Evolutionary Biology*, 28(5), 1080–1090.